

Course Code		Course Title				
316U06C107		Structured Programming Methodology				
Teaching Scheme (hrs.)	TH	PRACT	TUT	Total		
	02	02	--	04		
Credits Assigned		02	01	--	03	
Examination Scheme	Marks					
	CA		ESE (On-Screen*)	PR/OR Exam	LAB/TUT CA	Total
	IA	MSE				
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*The End Semester Examination (ESE) will be conducted in an onscreen format and may include a combination of theoretical questions, practical coding tasks, and/or an oral/viva component

Course prerequisites:

Students taking this course should have a basic understanding of computer operations, including familiarity with operating systems and file management. They are expected to possess fundamental logical reasoning skills and the ability to approach problems methodically. A basic grasp of mathematics, particularly arithmetic and simple algebra

Course Objectives:

The objective of course is to understand structured programming concepts and basic problem-solving techniques using algorithms, flowcharts, and pseudocode, implement decision-making and looping constructs to control program flow effectively, use arrays and strings for data storage and manipulation and apply functions, structures, and pointers to create modular and efficient programs.

Course Outcomes

At the end of the successful completion of the course, the student will be able to understand

CO1: Formulate a problem statement and develop the logic (algorithm/flowchart) for its solution

CO2: Demonstrate the use of control structures

CO3: Apply the concepts of arrays and strings

CO4: Design modular programs using functions and the use of structure

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction to Structured Programming methodology		05	CO1
	1.1	Problem solving skill development: Problem Definition, fundamentals of algorithms and flowcharts, Program Design, Pseudocode.		
	1.2	Structured Programming		
	1.3	Program execution process, Systems Development Life Cycle		
	1.4	Understanding concept and importance of header file/package/namespaces Data & Operators: Data Types, Identifier, Constants and Variables		
	1.5	Types of Operators, Expressions and Evaluation of Expressions, Operator Precedence and Associativity, Type Conversions		
2	Program Control Functions		06	CO2
	2.1	Decision Making and Branching Control Structures: Two Way Selection, Multiway Selection		
	2.2	Looping Control Structures, Flag Concept, Counting Loops		
	2.3	Documentation and Making Source Code Readable		
3	Introduction to Arrays		07	CO3
	3.1	Arrays: Introduction to One Dimensional Arrays, Multidimensional Arrays, Declaration and Initialization of Arrays, Reading and Displaying arrays		
	3.2	Character Arrays and Strings: Introduction, Declaring and Initializing String Variables, Reading Character and Writing Character, Reading and Writing Strings, various operation on strings, Implementation of string handling operations (from scratch)		
4	User defined function and Structures		12	CO4
	4.1	User Defined Functions: Need, Function Declaration and Definition, Return Values, Function Calls, Passing Arguments to a Function by Value, Recursive functions, String Handling Functions (inbuilt)		
	4.2	Structures and Unions: Introduction, Declaring and defining Structure, Structure Initialization, Accessing and Displaying Structure Members, Array of Structures		
	4.3	Introduction to pointers: Pointer declaration and initialization, Pointer addition and subtraction, evaluating pointer expressions Pointers and Functions: Pass by Reference, Returning pointers from functions		
	#	Self-Learning: Introduction to Unions, Structure Vs Unions, File Handling		
Total			30	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Kenneth Leroy Busbee	Programming Fundamentals - A Modular Structured	Rice University, Houston, Texas	2013

		Approach using C++		
2	Hassan Afyouni Behrouz A. Forouzan	Computer Science: A Structured Programming Approach Using C	Cengage India Private Limited	4 th Edition 2023
3	Behrouz A. Forouzan Richard F. Gilberg	Computer Science: A Structured Approach Using C++	Cengage India Private Limited	2nd edition 2012
4	E. Balagurusamy	Programming in ANSI C	McGraw-Hill Education, India	8 th Edition, 2019
5	Pradeep Dey and Manas Ghosh	Structured Programming Approach	Oxford University Press, India	1 st Edition, 2016
6	https://press.rebus.community/programmingfundamentals/chapter/structured-programming/			
7	https://tfetimes.com/wp-content/uploads/2015/04/structured-programming-with-c-plus-plus.pdf			
8	https://open.umn.edu/opentextbooks/textbooks/144			
9	NPTEL (C)	https://onlinecourses.nptel.ac.in/noc22_cs40/preview		
10	NPTEL (C++)	https://onlinecourses.nptel.ac.in/noc21_cs02/preview		